Patient Name:		Patient DOB:	
Date Collected:		Referring Physician:	
Biopsy Site:	**Bone marrow aspirate**	Primary Diagnosis:	**Acute myeloid leukemia**
Sample ID:		MoCha ID:	
Telephone:		Fax:	
Tumor Content (%):	**33%**		

Preanalytics to include tumor enrichment as required is performed by Van Andel Research Institute; Grand Rapids; MI 49503

Relevant Acute Myeloid Leukemia Findings

Gene	Finding	Gene	Finding
ABL1	None detected	MECOM	None detected
ASXL1	None detected	MLLT3	None detected
CEBPA	None detected	MYH11	None detected
CREBBP	None detected	NPM1	None detected
FLT3	None detected	NUP214	None detected
IDH1	None detected	RARA	None detected
IDH2	None detected	RUNX1	None detected
KMT2A	***KMT2A-ELL fusion***	TP53	None detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	*KMT2A-ELL fusion*	allogeneic stem cells azacitidine cytarabine cytarabine + daunorubicin cytarabine + daunorubicin + etoposide cytarabine + etoposide + idarubicin cytarabine + fludarabine + idarubicin + filgrastim cytarabine + idarubicin cytarabine + mitoxantrone decitabine gemtuzumab ozogamicin + chemotherapy venetoclax + chemotherapy	None	15

Prognostic significance: ELN 2017: Adverse

Public data sources included in relevant therapies: FDA1, NCCN
Public data sources included in prognostic and diagnostic significance: NCCN
Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Variant Details

Gene Fusions (RNA)

Genes	Variant ID	Locus
KMT2A-ELL	KMT2A-ELL.K9E2	chr11:118355029 - chr19:18583692

Low Coverage Regions:

N/A

Report Signed By

Name	Role	Date

Biomarker Descriptions

KMT2A (lysine methyltransferase 2A)

Background: The KMT2A gene encodes the lysine methyltransferase 2A protein, a transcriptional coactivator and histone H3 lysine 4 (H3K4) methyltransferase. KMT2A, also known as mixed lineage leukemia (MLL), is part of the SET domain protein methyltransferase superfamily. KMT2A influences epigenetic regulation by means of its methyltransferase activity which regulates a variety of cellular functions including neurogenesis, hematopoiesis, and osteogenesis[1]. Located at the chromosomal position 11q23, KMT2A is the target of recurrent chromosomal rearrangements observed in several leukemia subtypes including MLL, acute myeloid leukemia (AML), and acute lymphoblastic leukemia (ALL)[2]. Such translocations encode KMT2A fusion proteins that are oncogenic with simultaneous loss of KMT2A H3K4 methyltransferase activity[2]. Loss of methyltransferase activity along with partner gene gain of function contributes to increased HOX gene expression and promotes the transformation of hematopoietic cells into leukemic stem cells[2,3,4,5].

Alterations and prevalence: KMT2A fusions are observed in 3-10% of AML cases with highest frequencies in therapy-related AML (9%) and patients younger than 60 years (5%)[2,6,7]. KMT2A rearrangements including t(4;11)(q21;q23)/AFF1-KMT2A, t(9;11)(p22;q23)/MLLT3-KMT2A, t(11;19)(q23;p13.3)/KMT2A-MLLT1, t(10;11)(p12;q23)/MLLT10-KMT2A, and t(6;11)(q27;q23)/AFDN-KMT2A translocations account for about 80% of all KMT2A rearranged leukemias[2]. In infant acute leukemic cases, KMT2A rearrangement is reported in up to 70% of those diagnosed with either AML or ALL[2,8,9]. Mutations in KMT2A are also reported in diverse solid tumors including 10-20% of melanoma, stomach, bladder, and uterine cancers and around 5% of lung and head and neck cancers[10]. KMT2A alterations observed in solid tumors include nonsense or frameshift mutations which result in KMT2A truncation and loss of methyltransferase activity[10,11].

Potential relevance: KMT2A fusions are associated with variable prognosis based on the partner genes involved in the fusion[7]. For example, t(6;11)(q27;q23)/AFDN-KMT2A fusions are associated with poor prognosis whereas, t(9;11)(p22;q23)/MLLT3-KMT2A fusions confer more favorable or intermediate prognosis in AML[12,13,14]. Additionally, 11q23 rearrangements define an unfavorable karyotype in patients diagnosed with primary myelofibrosis (PMF) and may confer intermediate to high risk depending on concurrent cytogenetic abnormalities[15].

Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

KMT2A-ELL fusion

Relevant Therapy	FDA	NCCN	Clinical Trials*
Allogeneic hematopoietic stem cell transplantation	✖	●	✖
azacitidine	✖	●	✖
cytarabine	✖	●	✖
cytarabine + daunorubicin	✖	●	✖
cytarabine + daunorubicin + etoposide	✖	●	✖
cytarabine + etoposide + idarubicin	✖	●	✖
cytarabine + fludarabine + idarubicin + filgrastim	✖	●	✖
cytarabine + idarubicin	✖	●	✖
cytarabine + mitoxantrone	✖	●	✖
decitabine	✖	●	✖
gemtuzumab ozogamicin + cytarabine + daunorubicin	✖	●	✖
venetoclax + azacitidine	✖	●	✖
venetoclax + cytarabine	✖	●	✖
venetoclax + decitabine	✖	●	✖
allogeneic stem cells, chemotherapy	✖	✖	● (II)
chemotherapy, radiation therapy	✖	✖	● (II)
natural killer cell therapy, stem cell therapy	✖	✖	● (II)
DSP-5336	✖	✖	● (I/II)
KO-539	✖	✖	● (I/II)
SNDX-5613	✖	✖	● (I/II)
SNDX-5613 + venetoclax + chemotherapy	✖	✖	● (I/II)
stem cell therapy	✖	✖	● (I/II)
allogeneic stem cells, chemotherapy, radiation therapy	✖	✖	● (I)
BMF-219	✖	✖	● (I)
nintedanib, chemotherapy	✖	✖	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

● In this cancer type ○ In other cancer type ◐ In this cancer type and other cancer types ✖ No evidence

KMT2A-ELL fusion (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
pemigatinib, chemotherapy	✖	✖	● (I)
SNDX-5613, chemotherapy, steroid	✖	✖	● (I)
venetoclax, chemotherapy	✖	✖	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

KMT2A-ELL fusion

NCT ID	Title	Phase
NCT01760655	A Two Step Approach to Reduced Intensity Allogeneic Hematopoietic Stem Cell Transplantation for High Risk Hematologic Malignancies	II
NCT03121014	A Phase II Study of Intensity Modulated Total Marrow Irradiation (IM-TMI) in Addition to Fludarabine/ Busulfan Conditioning for Allogeneic Transplantation in High Risk AML and Myelodysplastic Syndromes	II
NCT02727803	Personalized NK Cell Therapy in Cord Blood Transplantation	II
NCT04988555	A Phase I/II, Open-Label, Dose-Escalation, Dose-Expansion Study of DSP-5336 in Adult Acute Leukemia Patients With and Without Mixed Lineage Leukemia (MLL)-Rearrangement or Nucleophosmin 1 (NPM1) Mutation	I/II
NCT04067336	A Phase I/IIa First in Human Study of the Menin-MLL(KMT2A) Inhibitor KO 539 in Patients with Relapsed or Refractory Acute Myeloid Leukemia	I/II
NCT04065399	AUGMENT-101: A Phase I/II, Open-label, Dose-Escalation and Dose-Expansion Cohort Study of SNDX 5613 in Patients With Relapsed/Refractory Leukemias, Including Those Harboring an MLL/KMT2A Gene Rearrangement or Nucleophosmin 1 (NPM1) Mutation	I/II
NCT03013998	A Master Protocol for Biomarker-Based Treatment of AML (The Beat AML Trial)	I/II
NCT01203722	Reduced Intensity, Partially HLA Mismatched Allogeneic BMT for Hematologic Malignancies Using Donors Other Than First-degree Relatives	I/II
NCT03494569	Phase I Study of Escalating Doses of Total Marrow and Lymphoid Irradiation (TMLI) Combined With Fludarabine and Melphalan as Conditioning for Allogeneic Hematopoietic Cell Transplantation in Patients With High-Risk Acute Leukemia or Myelodysplastic Syndrome	I
NCT05153330	A Phase I First-in-human Dose-escalation and Dose-expansion Study of BMF-219, an Oral Irreversible Menin Inhibitor, in Adult Patients With Acute Leukemia (AL), Diffuse Large B-cell Lymphoma (DLBCL), and Multiple Myeloma (MM).	I
NCT03513484	Phase Ib, Open Label, Combination Study of Nintedanib With 5-Azacitidine in Acute Myeloid Leukemia Characterized by HOX Gene Overexpression, That Are Not Candidates of Intensive Chemotherapy	I

Clinical Trials Summary (continued)

KMT2A-ELL fusion (continued)

NCT ID	Title	Phase
NCT04659616	A Multicenter, Phase I, Open-Label Study of the FGFR Inhibitor Pemigatinib (INCB054828) Administered After Chemotherapy in Newly Diagnosed Acute Myeloid Leukemia (AML) With Adverse or Intermediate Risk Cytogenetics	I
NCT05326516	AUGMENT-102: A Phase I, Open-label, Dose-escalation Study to Evaluate Safety, Tolerability and Preliminary Anti-Leukemic Activity of SNDX-5613 in Combination With Chemotherapy in Patients With Relapsed/Refractory Leukemias Harboring a KMT2A/MLL Gene Rearrangement or Nucleophosmin 1 Mutation (mNPM1)	I
NCT03613532	A Phase I Study of Adding Venetoclax to a Reduced Intensity Conditioning Regimen and to Maintenance in Combination With a Hypomethylating Agent After Allogeneic Hematopoietic Cell Transplantation for Patients With High Risk AML, MDS, and MDS/MPN Overlap Syndromes	I
NCT03844815	Phase I Study of Venetoclax in Combination With Decitabine 10-Day Regimen in Subjects With Acute Myeloid Leukemia	I

Assay Information

Methodology, Scope, and Application:The NCI Myeloid Assay v2 (NMAv2) is a next-generation sequencing (NGS) assay which identifies pre-defined and novel genomic variants covering 45 DNA genes and 35 fusion drivers that are categorized into 3 mutation types: single nucleotide variants (SNVs), insertions/deletions (Indels) including FLT3 internal tandem duplications (ITDs), and gene fusions. The assay utilizes Thermo Fisher Scientific's Oncomine® Myeloid Assay GX v2, a NGS assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from blood and bone marrow mononuclear cells for sequencing on the Ion Torrent Genexus Integrated Sequencer and analyzed by the Ion Torrent Genexus pipeline version 6.6.2.1. The NMAv2 can reliably identify the presence or absence of 1661 hotspot variants and 779 reported fusions compared to the Human Reference Genome hg19, including the genes listed below. The NMAv2 is a laboratory developed test designed to detect gene mutations for major myeloid disorders.

Analytical Sensitivity and Specificity:The NMAv2 has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 98.97% sensitivity, 100% specificity and an overall reproducibility with a mean positive percent agreement of 99% and a mean negative percent agreement of 100% during validation of the assay.

Hereditary and Germline Mutations:This assay examines cancer cells only and does not examine normal (non-cancer) cells. Mutations detected by the assay may be present only in the cancer cells, or in every cell of the body (including non-cancer cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or more features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's), it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-cancer) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content:Thermo Fisher Scientific's Ion Torrent Oncomine Knowledgebase Reporter (OKR) software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OKR. The data presented here is from a curated knowledgebase of publicly available information but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al.

2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:
ABL1, ANKRD26, BRAF, CBL, CSF3R, DDX41, DNMT3A, FLT3, GATA2, HRAS, IDH1, IDH2, JAK2, KIT, KRAS, MPL, MYD88, NPM1, NRAS, PPM1D, PTPN11, SETBP1, SF3B1, SMC1A, SMC3, SRSF2, U2AF1, WT1

Genes assayed with full exon coverage:
ASXL1, BCOR, CALR, CEBPA, ETV6, EZH2, IKZF1, NF1, PHF6, PRPF8, RB1, RUNX1, SH2B3, STAG2, TET2, TP53, ZRSR2

Genes assayed for detection of fusions:
ABL1, ABL2, BCL2, BRAF, CCND1, CREBBP, EGFR, ETV6, FGFR1, FGFR2, FUS, HMGA2, JAK2, KAT6A, KAT6B, KMT2A, KMT2A-PTD (Partial Tandem Duplication), MECOM, MET, MLLT10, MRTFA (MKL1), MYBL1, MYH11, NTRK2, NTRK3, NUP214, NUP98, PAX5, PDGFRA, PDGFRB, RARA, RUNX1, TCF3, TFE3, ZNF384

Limitaions of the assay

1. Mutations in homopolymer regions of >5 nucleotides may not be accurately assessed by this assay.
2. FLT3-ITDs greater than 120bps may not be detected by this assay. Sensitivity for larger FLT3-ITD’s may be reduced and variant allele frequencies may be reported lower than expected.
3. Following genetic alterations will not be reported by this assay: RUNX1 p.Ser445AlafsTer149, ASXL1 p.G646Wfs*12, PRPF8 p.Arg156GlufsTer28

DISCLAIMER

This assay reports SNV, indel and FLT3-ITD mutations ≥ 5% Variant Allele Frequency (VAF). This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

References

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